CATEGORY TREE CRUD FOR BOOK MANAGEMENT

C Data Structures Project

Team Members:
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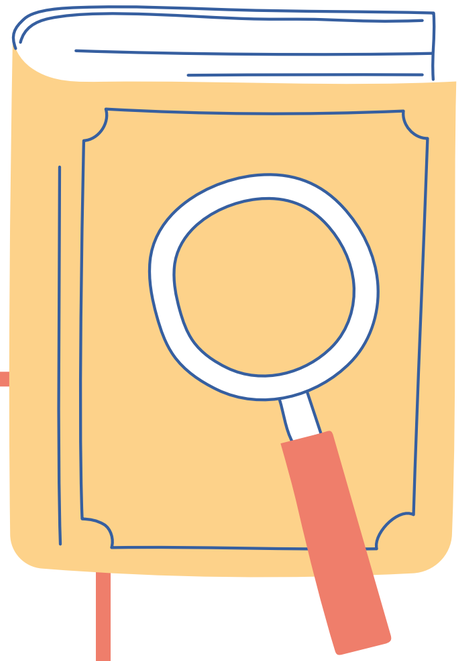
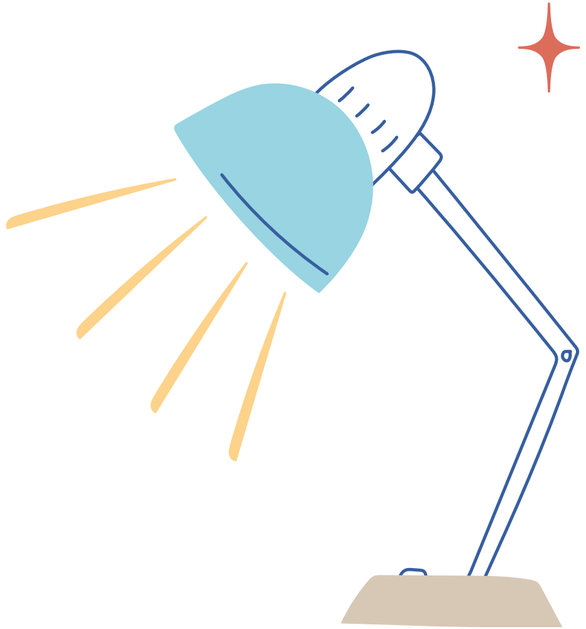
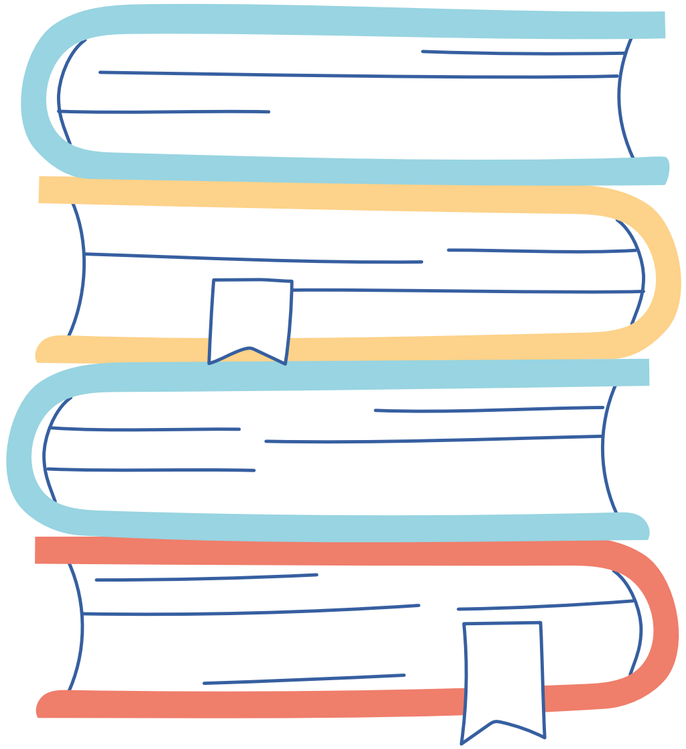
PROBLEM STATEMENT





- Online bookstores have many categories
- Categories are hierarchical (parent-child)
- Difficult to manage using simple structures
- Need efficient system for:
- Adding
- Updating
- Deleting categories



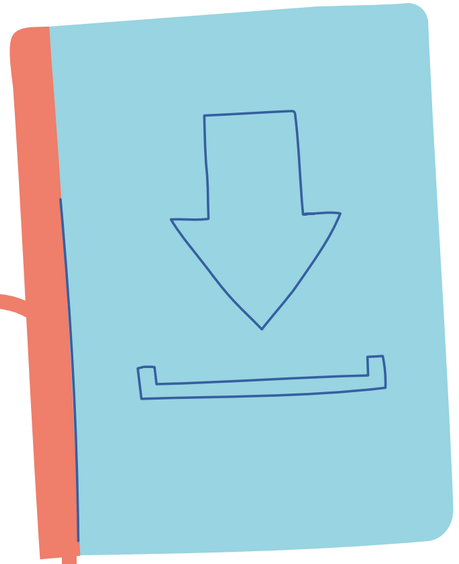
REAL-WORLD APPLICATION

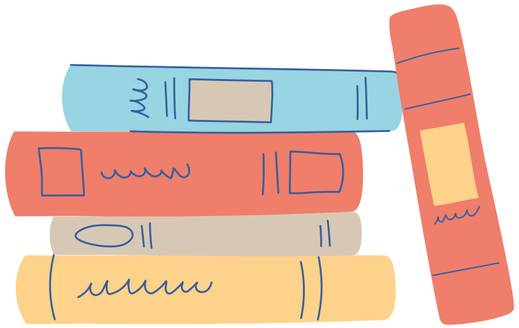
- Used in online bookstores
- Categories like:
- Fiction → Novel
- Education → Science
- Helps in:
- Easy navigation
- Organized data
- Better user experience

OBJECTIVE

- Implement tree data structure in C
- Perform CRUD operations
- Use dynamic memory allocation
- Develop menu-driven program
- Simulate real-world category system

DATA STRUCTURE USED





Tree (General Tree)

- Hierarchical structure
- Nodes represent categories
- Uses:
  - Child pointer
  - Sibling pointer

Example:

```
Books
├── Fiction
└── Education
```

SYSTEM DESIGN

- Each node contains:
  - Category name
  - Pointer to child
  - Pointer to sibling
- Flow:

User Input → Menu → Function → Tree → Output

ALGORITHMS

- Add Category → Insert node
- Search → Recursive traversal
- Update → Modify node value
- Delete → Remove node & free memory
- Display → Tree traversal

IMPLEMENTATION

- Language: C
- Concepts used:
- struct
- pointers
- malloc() and free()
- recursion
- Functions:
- createNode()
- addCategory()
- search()
- update()
- deleteNode()
- display()

OUTPUT / DEMO

- Menu-driven program
- Performs:
  - Add category
  - Search
  - Update
  - Delete
  - Display tree

Example Output:

- Books
- Fiction
- Novel

CONCLUSION

- Tree structure is efficient for hierarchical data
- CRUD operations successfully implemented
- Real-world application demonstrated

Future Scope:

- Add book details
- GUI application
- File/database storage